

Seasonal Employment: Replication and Extension

Coglianesse & Price (2025) + QWI 6-Digit NAICS Index

September 11, 2026

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Coglianese & Price (2025): Overview

- “Income in the Off-Season: Household Adaptation to Yearly Work Interruptions,” *ILRR* 2025
- **Question:** How do households smooth income when employment separations recur annually (i.e. are *seasonal*)?
- **Key innovation:** Identify seasonal separations from administrative data without requiring industry or occupation labels
- Core estimand: **excess recurrence** at 12-month horizon

$$\widehat{\text{ExR}} = \text{Pr}(\text{sep at } k=12) - \frac{\text{Pr}(\text{sep at } k=11) + \text{Pr}(\text{sep at } k=13)}{2}$$

- Counterfactual = average of adjacent lags $\rightarrow$ isolates the *annual spike* marking genuine seasonal separations

Event-study OLS on separators panel (CPS and SIPP):

$$\mathbf{1}[\text{sep at } k] = \sum_j \beta_j \mathbf{1}[k = j] + \varepsilon$$

- Estimate by OLS; excess recurrence = $\beta_{12} - (\beta_{11} + \beta_{13})/2$
- Standard error via delta method
- **Two specifications:**
 - ① *Raw* (Figure 2 in paper): only horizon dummies, no constant
 - ② *Preferred* (Appendix F.1): adds controls for interview-seam artifacts (SIPP) and weeks elapsed since separation (CPS)
- Python replication uses `statsmodels` WLS with clustered SEs; data from the LEHD-style pre-cleaned `.dta.gz` files in the replication package

Replication Results

Table: Excess recurrence estimates: paper vs. Python replication

Specification	Where	Paper	Replication
CPS raw (Figure 2)	Figure 2	1.4 p.p. (SE 0.14)	1.44 ✓
CPS preferred	App. F.1	1.5 p.p. (SE 0.14)	1.42 ✓
SIPP raw (Figure 2)	Figure 2	2.0 p.p. (SE 0.11)	2.04 ✓
SIPP preferred	App. F.1	1.6 p.p. (SE 0.11)	1.62 ✓

- All four estimates within rounding of published values
- Industry ranking: **Agriculture, Education, and Entertainment** are the top three by excess recurrence in *both* CPS and SIPP (order varies slightly by dataset)
- **Healthcare** is near the bottom in both; **Construction** is solidly mid-pack (8th/15 CPS, 7th/15 SIPP) — not top-tier, despite common intuition. (Mining flips sign across datasets on a small sample and is excluded from ranking claims.)

Extending to 6-Digit NAICS via QWI

- CP's classifier operates at the worker level; industry rankings are reported at 1-digit NAICS only
- **Goal:** produce a finer, industry-level seasonal index suitable for firm classification
- **Data:** Census Quarterly Workforce Indicators (QWI)
 - Quarterly separations and employment by state $\times$ 6-digit NAICS
 - 51 states + DC, years 2000–2023 (1,011 industries)
- **Quarterly analog to excess recurrence:** for each quarter $q \in \{1, 2, 3, 4\}$ and industry j ,

$$\text{Excess}(q)_j = \text{SepRate}_{j,q,t} - \frac{\text{SepRate}_{j,q-1,t} + \text{SepRate}_{j,q+1,t}}{2}$$

(quarters treated cyclically; average over years t)

- **Peak excess:** $\text{ExcessPeak}_j = \max_q \text{Excess}(q)_j$
- **Seasonal amplitude:** $\max_q - \min_q$ normalised to $[0, 1]$; does not assume any particular quarter is “the” seasonal quarter

National Seasonal Index: Top Industries

Table: Top 12 most seasonal 6-digit NAICS industries (national)

NAICS & description	ExcessPeak	n
111336 Fruit/tree nut combo farm.	0.333	18
722330 Mobile food services	0.316	18
111140 Wheat farming	0.287	20
111160 Rice farming	0.563	22
111199 Other grain farming	0.398	23
111219 Other veg. (ex. potato) farm.	0.462	16
111998 Other misc. crop farming	0.328	20
111120 Oilseed (ex. soybean) farm.	0.348	12
115113 Crop harvesting (machine)	0.398	10
512132 Drive-in motion picture theaters	0.363	21
532284 Recreational goods rental	0.485	22
111191 Oilseed/grain combo farm.	0.321	23

- n = years contributing to ExcessPeak estimate; a ≥ 5 -year minimum is enforced so thin, disclosure-suppressed cells can't masquerade as extreme seasonality
- Non-agriculture entries are intuitive once named: mobile food services (food trucks), drive-in theaters, recreational goods rental
- Seasonal index normalized to $[0,1]$ via 99th-pct winsorization
- Correlation with CP 1-digit rankings: $\rho = 0.829$

National Seasonal Index: Sector Summary

Table: Mean seasonal index by 2-digit sector (national)

Sector	Mean seasonal index	ExcessPeak (p.p.)
Agriculture, forestry, fishing	0.504	15.7
Arts, entertainment & recreation	0.461	14.3
Accommodation & food services	0.345	10.3
Educational services	0.272	8.4
Construction	0.193	5.9
Real estate	0.167	5.6
Warehousing & couriers	0.135	4.6
Other services	0.134	4.2
Health care & social assistance	0.067	2.2
Wholesale trade	0.063	1.8
Manufacturing (metals/machinery)	0.036	1.1

Education peaks in Q2 (end of school year), not Q4 — ExcessPeak is positive across all sectors. CP's 1-digit rankings confirmed at finer granularity.

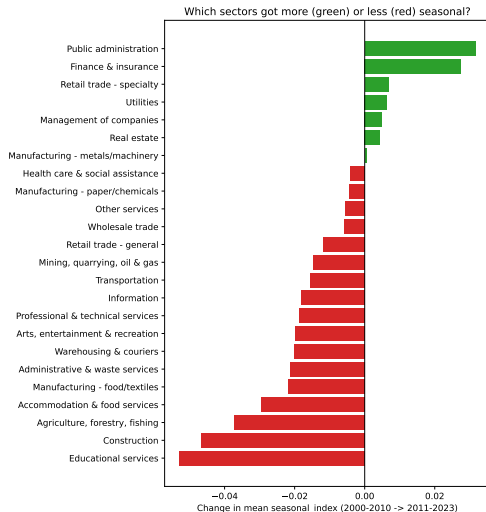
Robustness Check: Does COVID Distort the Index?

- The national index pools 2000–2023. Mass pandemic-era layoffs in industries already ranked highly seasonal (accommodation/food, arts & entertainment) could inflate “excess separation” for reasons that have nothing to do with normal annual seasonality
- **Test:** recompute the index restricted to 2000–2019 only, compare to the full sample
- **Result: rankings are robust.**
 - Pearson correlation (full vs. pre-COVID seasonal index): 0.997
 - Spearman rank correlation: 0.986
 - Only 6.7% of industries have a different peak quarter
 - Watch sectors (agriculture, construction, education, arts/entertainment, accommodation/food) all shift by ≤ 0.017
- **One genuine mover:** a transportation industry (transit, NAICS 485111) whose index is roughly $4.7\times$ higher with COVID years included — plausibly a real ridership-collapse effect, not a data artifact

Seasonality Over Time: 2000–2010 vs. 2011–2023

- The national index pools all 24 years. Has seasonality actually been stable, or has it drifted over two decades?
- **Test:** split the sample in two and recompute independently. The 5-year *n_excess_obs* minimum is what makes a clean two-way split possible — both halves (11 and 13 years) keep near-full coverage, so a three- or four-way split isn't advisable
- **Result: correlated, with real drift.**
 - Pearson $r = 0.931$, Spearman $r = 0.826$ (both notably lower than the COVID check's 0.997/0.986 — an 11-year gap between period midpoints is a much bigger ask than excluding 2 years)
 - Economy-wide mean seasonal index is essentially flat: $0.131 \rightarrow 0.122$
- **But the flat average hides a real reshuffling** beneath it (next slide)
 - 30.2% of industries have a different peak quarter, but this concentrates in weakly-seasonal industries (43% of the bottom quartile vs. 15% of the top) — ordinary noise, not genuine timing instability among strongly seasonal industries

Which Sectors Got More or Less Seasonal?



Educational services: largest decline

- -0.053 , the biggest sector decrease ($0.272 \rightarrow 0.219$)

Construction: also declining

- -0.047 ($0.216 \rightarrow 0.169$)

Agriculture: modest decline

- -0.037 ($0.504 \rightarrow 0.466$)

Finance/insurance and public admin: rose most

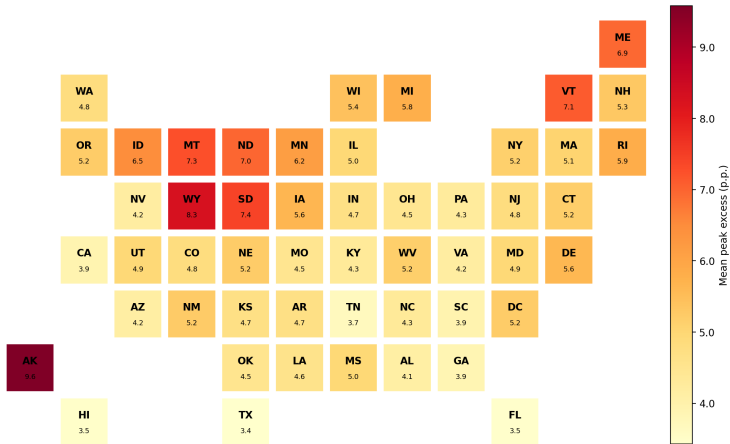
- $+0.028$ and $+0.032$ respectively — the only two sectors with a meaningful increase

Geographic Variation: Overview

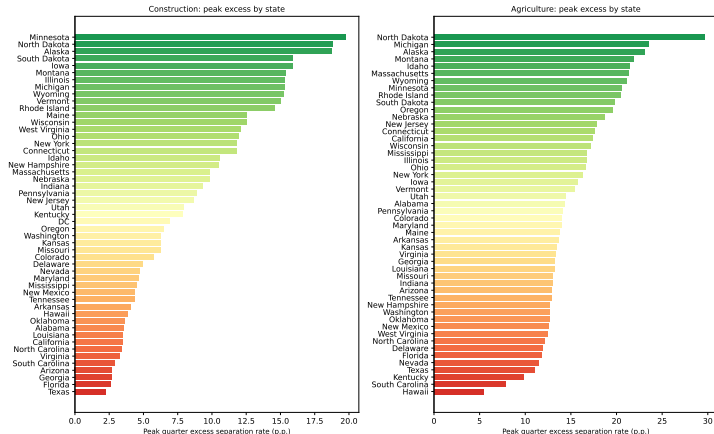
- The national index masks substantial state-level heterogeneity
- We compute ExcessPeak for each state $\times$ 6-digit NAICS cell (42,344 cells with ≥ 5 year-observations)
- **Main finding:** a $2.8\times$ gap between most and least seasonal states in mean peak excess (Alaska: 9.6 p.p. vs. Texas: 3.4 p.p.)
- Northern/Great Plains states consistently more seasonal across multiple sectors; Sun Belt states consistently less so
- **Most geographically variable sectors:**
 - ① Agriculture (cross-state SD = 14.5 p.p.)
 - ② Arts & entertainment (SD = 13.4 p.p.)
 - ③ Accommodation/food (SD = 10.8 p.p.)
 - ④ Construction (SD = 9.6 p.p.)
- **Least geographically variable:** healthcare, metals manufacturing, utilities — consistent low seasonality everywhere

State-Level Mean Seasonality

State-level seasonality
Mean peak excess separation rate across 6-digit NAICS industries



Construction and Agriculture by State



Construction: Minnesota (19.7 p.p.) vs. Florida (2.6 p.p.) — an 8× gap driven by winter construction shutdowns in northern states. **Agriculture:** North Dakota (29.6 p.p.) leads, ahead of Michigan (23.5 p.p.), Alaska (23.1 p.p.), Montana, and Idaho.

State-Specific Top and Bottom 1% Industries

Table: Within-state NAICS6 tails

Statistic	Top 1%	Bottom 1%
State-industry slots	413	413
Unique NAICS6 codes	137	253
Codes appearing once	59	156
Codes in 10+ states	7	0
Pairwise Jaccard	0.053	0.014
Also national same tail	11.4%	1.9%

Interpretation: the state-specific tails are related to the national index, but most state-tail industries are not simply the national top or bottom 1% repeated everywhere.

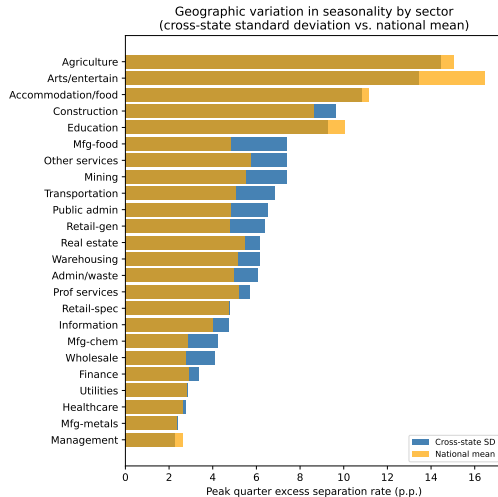
Recurring top-1% industries

- Nursery/tree production: 15 states
- Marinas: 14 states
- Specialty trade contractors: 12 states
- Golf courses/country clubs: 11 states
- Miscellaneous crop farming: 11 states

Bottom-1% pattern

- More diffuse than the top tail
- Concentrated in metals manufacturing, wholesale, chemicals, finance, and transportation

Geographic Variation: Sector Cross-State SD



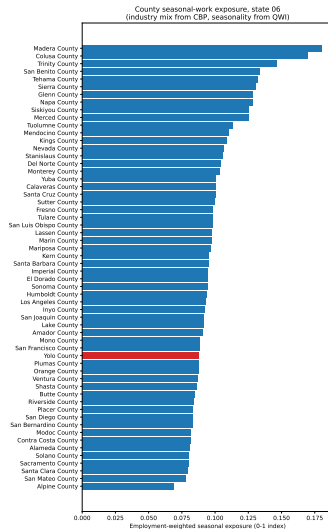
County-Level Seasonal Exposure: Methodology

- **Goal:** how seasonal is the average job in a given county, given the industries actually located there?

$$\text{Exposure}_c = \frac{\sum_j \text{EMP}_{c,j} \times \text{seasonal_index}_j}{\sum_j \text{EMP}_{c,j}}$$

- **Data:** County Business Patterns (CBP) establishment/employment counts by county $\times$ NAICS, merged onto the state $\times$ NAICS6 seasonal index
 - **Problem:** CBP suppresses most county $\times$ NAICS6 cells (disclosure avoidance) — a county's genuinely seasonal niche industries are often exactly the small, easily-identified cells that get suppressed
 - **Fix: three-tier fallback** for each industry's seasonal score: (1) state $\times$ NAICS6, (2) national NAICS6, (3) national NAICS4 (also recovers employment CBP suppresses at 6-digit but still reports at 4-digit)
- California pilot: **100% employment coverage** in all 58 counties — no county's score rests on a partial industry match

California County Ranking



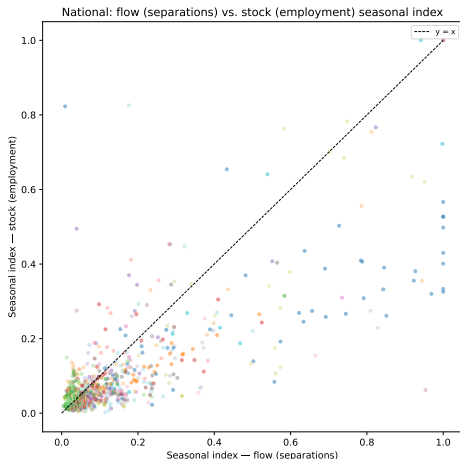
Case Study: Yolo County

- **Yolo County ranks 41st of 58** California counties (30th percentile) — *less* exposed to seasonal work than roughly two-thirds of the state, despite being an agricultural county
 - **Why:** employment-weighting means large, stable service/logistics employers dominate the score:
 - Limited-service restaurants (3,222 jobs, index 0.10), general warehousing (3,087 jobs, index 0.06), couriers (1,731 jobs, index 0.14) — all low-seasonality but large employers
 - Crop harvesting scores the *maximum* 1.00 — but employs only 246 people locally
 - **Context:** Central Valley farm counties top the ranking (Madera 0.180, Colusa 0.169 — roughly 2× Yolo's 0.088); the rest of the Sacramento metro area (Sacramento, Solano, Placer) clusters near Yolo at the low-seasonality end
- A county can have extreme seasonal niches and still score low overall, if its broader economy has diversified around them

Flow vs. Stock: Motivation

- The index is built two ways in parallel:
 - **Flow** (`seasonal_index`): from separation *rates* — does a quarter see an excess of separations?
 - **Stock** (`seasonal_index_emp`): from employment *levels* normalized to their own annual mean — does headcount itself swell or shrink that quarter?
 - **Result: correlated.**
 - National ($n = 997$): Pearson $r = 0.784$, Spearman $r = 0.660$
 - State $\times$ NAICS6 ($n = 39,446$): Pearson $r = 0.750$, Spearman $r = 0.661$
 - **Peak-quarter concordance is low** — **$\sim 15\%$ nationally, $\sim 13\%$ by state**: even when both measures agree an industry is seasonal, they usually disagree on *which quarter* drives it
- Timing, not just amplitude, differs between “separations spike” and “headcount swings” (see next slide)

Flow vs. Stock: Sector Heterogeneity



Construction: near-perfect agreement

- Flow/stock correlation $r = 0.97$ ($n = 31$):
winter layoffs = winter headcount drop

Agriculture: weaker, not decoupled

- $r = 0.64$ ($n = 55$) — meaningfully positive,
though less tightly linked than Construction

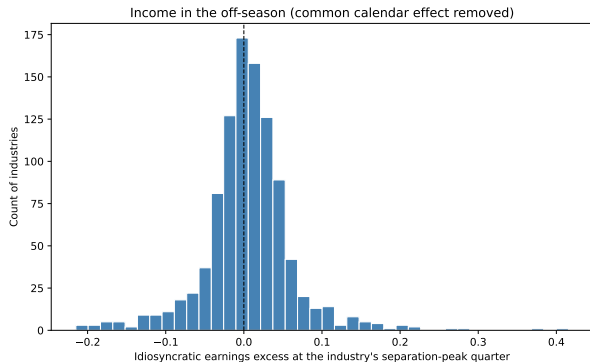
“Stock without flow” examples

- Cotton Ginning (flow 0.01, stock 0.82) pairs
with Political Organizations (flow 0.18, stock
0.83) — hiring surges (harvest; elections)
without a matching separation spike

A Third Lens: Does Income Dip in the Off-Season?

- Flow and stock both measure separations/headcount. A third QWI measure, EarnBeg (average monthly earnings), tests the paper's title question directly: **does income dip in an industry's own off-season?**
 - *Data note:* the originally-requested Payroll field returns null from the Census API for every cell tested (even huge industries where employment populates fine); EarnBeg is the working substitute
 - **A confound has to be fixed first:** 83% of *all* 1,011 industries — not just seasonal ones — have their raw earnings index peak in Q4. That's a broad, economy-wide year-end bonus/holiday-pay effect, not industry-specific timing
 - Left uncorrected, this swamps the signal: whether an industry's *own* separation-peak quarter happens to land in Q4 (everyone's bonus quarter) almost entirely determines whether it looks like income "dips" there
- **Fix:** compute the common cross-industry calendar effect (Q1 -0.031 , Q2 -0.005 , Q3 -0.048 , Q4 $+0.085$) and net it out before testing — the *idiosyncratic* index isolates industry-specific timing

Does Income Dip in the Off-Season? The Answer



Naive test: 52.3% "dip" — but almost entirely mechanical:

- Only 8.3% dip when the separation peak falls in Q4 (everyone's earnings peak)
- 93.6% "dip" when it falls in Q3 (rarely anyone's earnings peak)

De-confounded (credible) test:

- **44.2% dip**, mean excess $+0.0055$ — essentially flat, not clearly negative
- No strong evidence of industry-average income dipping in the off-season

Construction: the sharp counter-example

- Dip rate only 3.2%, mean excess $+0.044$ — earnings *higher*, plausibly overtime pay before winter layoffs

Doesn't contradict CP: their effect is the separating worker's own income loss (micro); this tests the industry-wide average (macro) — a different, complementary question.

Same Pay, Opposite Seasonality: Does It Matter?

- **Question:** among female-dominated industries, does seasonal work leave otherwise-comparable workers more rent-burdened?
- **Pay:** ACS PUMS WAGP — wage/salary income a person *actually received* in the past 12 months, not a QWI pay *rate* annualized $\times 12$. This matters specifically for seasonal work: someone who doesn't work all four quarters reports their true (lower) realized income, not a year-round-equivalent rate
- **Gender & pay & seasonality, one resolution:** ACS/CPS survey data classifies industries more coarsely than 6-digit NAICS (e.g. Educational Support Services is grouped with five sibling “other schools and instruction” codes). Pay, %female, and QWI seasonality (aggregated up from NAICS6, employment-weighted) are all computed at that same resolution, so the three are directly comparable
- **Rent benchmark:** Census ACS national median gross rent, \$1,413/month
- **Screen:** female-dominated ($\geq 55\%$), within 15% on pay, seasonal-index gap ≥ 0.25 — only 9 of 250 industry codes clear the female-share and sample-size bars

Matched Pairs: Same Pay, Opposite Seasonality

Table: Same-pay pairs, one seasonal and one flat (both female-dominated)

Industry (Census industry code)	Seasonal idx.	Pay	Rent burden
Ed. support & other schools (peak Q2)	0.51	\$20,000	84.8%
Other personal services (flat)	0.13	\$20,000	84.8%
Civic/social/advocacy orgs (peak Q3)	0.43	\$50,000	33.9%
Outpatient care centers (flat)	0.04	\$48,000	35.3%
Civic/social/advocacy orgs (peak Q3)	0.43	\$50,000	33.9%
Other health care services (flat)	0.06	\$50,000	33.9%

- **Top pair:** both sides *exactly* \$20,000 realized median pay, 65%/68% female — both **84.8% rent-burdened**
- **Weeks worked doesn't explain the pay level:** every pair differs by <1 week/year (~46–49 either way). The pay gap tracks each industry's wage level, not annual weeks logged — closer to CP's household-adaptation story (workers find other work around a seasonal separation) than “seasonal → fewer weeks”
- **Caveat:** WAGP is a personal wage — many \$20,000 earners are secondary earners or part-time, not sole rent-payers

Implications for Firm Classification

- The QWI index enables firm-level seasonal classification via NAICS lookup
- The top/bottom 1% analysis shows the exact NAICS6 tails differ sharply across states: top tail has 137 unique codes; bottom tail has 253
- **But:** for geographically sensitive sectors (construction, agriculture), a single national index may misclassify:
 - A Minnesota construction firm is highly seasonal
 - A Florida construction firm is nearly non-seasonal
- **Preferred approach for firm classification:**
 - ① Use the *state* $\times$ *NAICS* ExcessPeak as the lookup value when state of operation is known
 - ② Fall back to national index for industries with thin state-level coverage (low *n_excess_obs*)
 - ③ Direct estimation from UI wage records when sufficient separations are available ($n \geq 20$ per firm)
- **Which index?** Flow suits *turnover risk*; stock suits *headcount/staffing swings* — correlated ($r \approx 0.78$) but timing (peak quarter) still agrees only $\sim 15\%$ of the time
- **Caveat:** 6-digit cells are often suppressed in small states; state-level estimates for niche agriculture sub-industries may rest on very few year-observations

Summary

- 1 **Replication:** Python replication of CP (2025) excess recurrence matches all four published estimates within rounding
- 2 **National index:** QWI-based ExcessPeak at 6-digit NAICS correlates strongly with CP's 1-digit rankings ($\rho = 0.829$); robust to excluding 2020–2021 (Pearson 0.997). Economy-wide seasonality is roughly flat 2000–10 to 2011–23, but Educational services fell most (-0.053), Construction also fell (-0.047), and Agriculture fell modestly (-0.037); Finance/insurance and public admin rose most ($+0.028$, $+0.032$)
- 3 **Geographic variation:** large state heterogeneity ($2.8\times$ gap, Alaska vs. Texas); a California county-level pilot applies the same logic down to counties — Yolo County ranks 41st of 58 CA counties, less seasonal than most despite its agricultural economy
- 4 **Flow vs. stock:** separations-based and employment-based seasonality are correlated ($r \approx 0.78$ national) but timing concordance is low ($\sim 15\%$); Construction near-identical ($r = 0.97$), Agriculture moderate ($r = 0.64$)
- 5 **Income/earnings:** a third lens (EarnBeg) finds no strong evidence of industry-average income dipping in the off-season once a common Q4-bonus effect is removed; Construction again the outlier (earnings *higher*, not lower)
- 6 **Pay, gender, and rent burden:** matching female-dominated industries on realized annual pay (ACS PUMS), the most seasonal vs. flattest pair — both at exactly \$20,000/year, 65–68% female — are *equally* rent-burdened (84.8% each) at the national median rent; weeks worked is nearly identical across every matched pair, so the pay level tracks the industry, not seasonality itself
- 7 **Next steps:** national CBP rollout for a full county-level index; apply the state $\times$ NAICS index to UI wage records for firm-level classification